Yalamanchili Harsha Vardhan
Computer Science & Engineering
Indian Institute of Technology Bombay

24B1069
B.Tech.
Gender: Male
DOB: 03/01/2007

Examination	University	Institute	Year	CPI / % Credits
Graduation	IIT Bombay	IIT Bombay	2028	8.15 **Core:** 128 **| Total:** 134
Intermediate	CBSE	Narayana English Medium School	2024	97.80%
Matriculation	CBSE	Narayana High School	2022	97.60%

Scholastic Achievements

- Offered admission with a merit **fellowship** (full tuition + stipend) to **IISc Bangalore** B.Tech Math & Computing, one of only **50** seats spanning AI/ML, theoretical CS, and quantum computing; chose **IIT Bombay CSE** *(2024)*
- Advanced to the **IIIT-Hyderabad UGEE CS research** interview shortlist (top **2%** of 75,000+ national candidates) by clearing simultaneous **SUPR** and **REAP** cutoffs screening for subject proficiency and research aptitude *(2024)*

Experience

Ocean Transworld Logistics | *Software Developer Intern, Contract* *(May - June 2026)*
Team of contract SWE interns; owned 10 of 15 feature modules

- Built and deployed the **CRM**'s enquiry-to-quote pipeline: a Python IMAP sidecar polling every 60s, **Node.js** API, WAL-mode **SQLite** with 13 tables, and a **React 19** SPA behind **Nginx**/TLS on a **GCP** VM under PM2
- Wrote a 5-signal **RFQ scorer** that ranks incoming emails by sender, keywords, attachments, and thread position, with regex and PDF/Excel parsing over **IMAP/SMTP**; used **message-ID** keys to avoid duplicate ingestion
- Implemented the freight logic: **L1 rate freeze**, a **margin calculator** with a configurable threshold lock requiring approval below 12%, **agent dispatch** with IMAP reply threading, and an append-only **audit log** for every enquiry

Research Experience

Theoretical Cryptography | *Research Project* *(June 2026 - Present)*
Guide: Prof. Sruthi Shekar - IIT Bombay

- Investigating **pseudorandom codes** (PRCs), error-correcting codes that look random to any efficient adversary, and the public-key constructions from **LPN**/**LWE** hardness used for **AI watermarking** of generative model outputs
- Reading up on the **Christ–Gunn** framework ahead of **R&D** this semester: how its decoding trapdoors give standard-model **CCA** security, and why adding error-correction structure makes a code harder to keep **pseudorandom**

Key Projects

Graph-Based Routing System | *Course Project* *(Aug - Nov 2025)*
Guide: Prof. Ashutosh Gupta - Data Structures and Algorithms; team of 3

- Built a **3-phase routing system** in **C++17** modeling a real road network: **Dijkstra** in dual distance/time modes, **KNN** in Euclidean and shortest-path variants, and dynamic edge add/remove with **JSON** query support
- Implemented **Yen's k-shortest paths** with **bidirectional A*** and **Euclidean heuristic** to **5,000+** nodes
- Engineered a **TSP-style delivery scheduler** with **K-means** driver clustering, priority rebalancing, and accident simulation over multi-driver dispatch at ∼60 km/h, modeling 5% cancellations and 10% priority orders

AI Chess Engine | *Self Project* *(June - July 2025)*

- Built a **C++ chess engine** on a **magic-bitboard** move generator with a hand-tuned evaluation combining **piece-square tables**, **phase-dependent king safety**, and separate **midgame** and **endgame** evaluation tables
- Integrated **Polyglot opening book** and **alpha-beta pruned minimax**, defeating a **1500-rated** Lichess bot

xv6 Kernel & Concurrency | *Course Project* *(Jan - May 2026)*
Guide: Prof. Purushottam Kulkarni - Operating Systems

- Extended the **xv6 kernel** with priority scheduling, `clone()` threading, lazy paging, and shared-memory IPC
- Built **semaphores** and **futex**-like primitives in **xv6**, plus a **pthread** scheduler with thread-safe queues
- Developed an **interactive C shell** supporting background execution, pipes, I/O redirection, and **POSIX signals**
- Designed persistent access control (sudo/unsudo) and per-file R/W permissions in the xv6 **inode-based file system**

Applied Crypto & Exploits | *Course Project* *(Aug - Sept 2025)*
Guide: Prof. Sruthi Shekar - Introduction to Cryptography

- Mounted a **timing side-channel** attack on **HMAC verification**, recovering a 10-char MAC byte-by-byte by amplifying **1s/byte** timing differences; also exploited **CTR-mode** keystream reuse to recover a secret flag
- Implemented the **Nostradamus diamond** collision for **Merkle–Damgård** hashes: precomputed a parallelisable diamond enabling chosen-target forced-prefix (**CTFP**) messages, with 100% success over 100 forging trials

Other Projects

Advanced Deep Learning & Sequence Modeling | *Course Project* *(Jan 2026 - May 2026)*
Guide: Prof. Sunita Sarawagi - Artificial Intelligence and Machine Learning

- Implemented **Encoder-Decoder RNNs** and **Block-Recurrent Transformers** from scratch in **PyTorch**, including **multi-head scaled dot-product attention** with symmetric and windowed variants for long-sequence modeling
- Built **1D/2D CNNs** with **residual connections** and **GAP** for motif detection and image classification
- Implemented a **Mixture of Experts** with **soft-routing**, and **SVMs** via **SMO** with learnable **RBF kernels**

Competitive Programming | *Summer of Science* *(June - July 2025)*
Maths and Physics Club, IIT Bombay

- **Codeforces Expert** (rated **1620**); solved **300+ problems** spanning **dynamic programming**, greedy, graphs (**Dijkstra**, **Floyd–Warshall**), **number theory**, bit manipulation, and string algorithms across difficulty tiers
- Applied **segment trees**, **Fenwick trees**, **union-find**, and **sparse tables** to meet tight time and memory limits

Summer of Quant | *Quant Community, IIT Bombay* *(June - July 2025)*

- Built a **Python backtesting engine** over 1,577 daily BTC candles (2019–2024) with a **brokerage-fee model** and next-candle execution to avoid lookahead, reporting **Sharpe** and drawdown against a **buy-and-hold** benchmark
- Designed a **five-state signal protocol** (open, reverse, hold) driven by **ATR** and volume-spike indicators, with configurable position sizing across both fixed-stake and profit-compounding modes on $1,000 initial capital
- Awarded a **Certificate of Excellence** as a **top-10** finisher among 500 participants in **Summer of Quant**

SAT-Based Sokoban Solver | *Course Project* *(Aug 2025)*
Guide: Prof. S. Krishna - Logic in Computer Science

- Encoded Sokoban mechanics into **Boolean satisfiability** using **PySAT** (**glucose3**), modeling **6+ constraint types** — movement logic, collision dynamics, non-overlap, and goal satisfaction — across discrete time steps
- Enhanced solver via **unit propagation** and **clause learning**, optimizing variable encoding for **SAT** scalability; reused the framework for a **729-variable Sudoku** encoding across **9×9** grids with row, column and block uniqueness

Data Analysis & Interpretation | *Course Project* *(Aug - Nov 2025)*
Guide: Prof. Ajit Rajwade - Data Analysis and Interpretation

- Built a facial-recognition system using **PCA** eigenfaces via **Gram-matrix** eigen-decomposition for high-dimensional template matching, and implemented **kernel density estimation** with cross-validated bandwidth selection
- Applied **Kolmogorov–Smirnov** tests, **cross-correlation**, and **maximum likelihood estimation** for distribution analysis, medical image registration to align noisy MRI scans, and robust plane-fitting on **3D point clouds**

Python Tools & Utilities | *Self Projects* *(2024 - Present)*

- Built a **Sudoku solver** that extracts grids from photos via **OpenCV** (perspective correction, contour detection, segmentation) and classifies digits with a **CNN**, then solves the board through constraint-propagation **backtracking**
- Wrote utilities: a **Codeforces** progress tracker, a Windows app-opacity tool, and a dark-mode PDF converter

Technical Skills

Languages	C/C++, Python, Bash, Assembly, Verilog, JavaScript, SQL, HTML/CSS
Software & Tools	Git, GitHub, Make, LaTeX, Markdown, Sed, Awk, Jupyter, ReactJS, Nginx, PM2
Crypto/Security	PyCryptodome, Hash Functions, Side-Channel Analysis, Constant-Time Programming
Data/ML Libraries	NumPy, Pandas, Matplotlib, SciPy, PyTorch, OpenCV, Scikit-learn
Systems & Infra	Linux, GCP, SQLite, REST APIs, IMAP/SMTP, Multithreading, Concurrency

Position of Responsibility

Student Mentor | *Summer of Science '26 | MnP Club, IIT Bombay* *(May - July 2026)*

- Developed **problem sets** and supplementary materials covering **cryptography** and **zero-knowledge proofs**
- Mentored **10 students** through weekly **tutorials**, problem-solving exercises, and one-on-one doubt resolution

Courses Undertaken

Computer Science	†Data Structures & Algorithms, Discrete Structures, Design and Analysis of Algorithms, †Digital Logic Design and Computer Architecture, Logic for Computer Science, †Operating Systems, †Software Systems Lab, Introduction to Cryptography, †Abstractions and Paradigms* for Programming, †Computer Networks*, Automata Theory*
Data Science	†Artificial Intelligence and Machine Learning, Data Analysis and Interpretation
Mathematics	Calculus, Linear Algebra & Differential Equations
Others	Makerspace, Biology, Entrepreneurship, Philosophy, Chemistry Lab, Physics Lab, Economics

†*Course has corresponding lab* **To be completed by Dec 2026*

Extracurricular Activities

- Built a fully operational **quadcopter** from scratch — structural design, electronics, and embedded firmware — with parts modeled in **Fusion 360** and a **wireless flight controller** for stable orientation and movement control
- Volunteered year-long with **NSS** (Green Campus), supporting environmental and campus sustainability initiatives